

CHUPUNGNYEONG, Korea, Republic of

WMO: 471350

Lat: 36.2203N

Lon: 127.9944E

Elev: 246

StdP: 98.41

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.4	-8.6	-20.2	0.6	-5.9	-18.3	0.8	-4.6	11.2	-3.2	9.9	-4.1	4.1	270	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	32.0	24.0	30.7	23.4	29.4	22.6	25.1	29.8	24.5	29.0	24.0	28.0	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.3	27.5	23.2	18.5	26.9	22.7	17.9	26.4	78.3	30.0	75.9	29.1	73.5	28.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.4	6.5	-13.2	33.7	2.1	1.1	-14.7	34.5	-15.9	35.2	-17.1	35.8	-18.6	36.7		
			-14.2	26.2	1.9	0.8	-15.6	26.8	-16.8	27.2	-17.9	27.7	-19.3	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-1.7	0.7	5.7	12.0	17.2	21.2	24.4	24.7	19.7	13.7	6.9	0.3
	DBStd	9.70	3.53	3.67	3.68	3.62	2.63	2.30	2.59	2.55	2.77	3.19	3.83	3.72
	HDD10.0	1196	363	262	139	19	0	0	0	0	7	106	299	
	HDD18.3	2818	622	494	390	192	53	4	1	0	17	146	342	558
	CDD10.0	1973	0	1	7	79	223	336	446	455	290	122	14	0
	CDD18.3	554	0	0	0	2	18	91	188	197	57	2	0	0
	CDH23.3	4049	0	0	1	38	221	604	1386	1498	291	11	0	0
	CDH26.7	1159	0	0	0	3	32	130	448	501	46	0	0	0
Wind	WSAvg	2.6	3.6	3.3	3.3	3.1	2.6	2.0	1.8	1.7	1.7	2.0	2.6	3.2
Precipitation	PrecAvg	1146	24	34	57	83	84	141	258	216	130	49	49	24
	PrecMax	1606	86	105	169	242	193	464	541	472	378	134	141	60
	PrecMin	762	1	1	8	22	12	10	80	48	4	1	11	1
	PrecStd	222	21	27	33	53	46	88	110	98	102	36	32	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	14.3	20.7	26.6	29.3	31.3	33.6	33.7	29.9	24.8	20.2	12.2
		MCWB	5.6	9.2	11.1	15.2	16.5	20.2	24.9	24.4	22.5	16.3	13.3	7.4
	2%	DB	7.2	11.4	17.5	24.3	27.4	29.5	32.0	32.3	28.0	22.7	17.4	9.8
		MCWB	3.3	5.8	8.9	13.7	16.2	19.8	24.4	24.2	21.3	15.9	11.6	5.5
	5%	DB	5.3	9.0	15.2	22.1	25.8	28.2	30.7	31.0	26.4	21.1	15.3	7.9
		MCWB	1.7	4.1	7.7	12.4	15.5	19.4	24.0	23.8	20.3	15.1	10.5	4.3
	10%	DB	3.6	6.9	12.9	19.8	24.2	26.8	29.3	29.6	25.0	19.6	13.3	6.2
		MCWB	0.6	2.6	6.8	11.3	15.0	19.0	23.5	23.6	19.6	14.4	9.4	3.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	11.1	12.4	18.0	20.0	23.2	25.9	26.2	24.0	19.4	15.2	8.4
		MCDB	8.5	13.5	16.4	22.4	23.4	27.4	31.2	31.1	27.7	21.9	18.2	11.3
	2%	WB	3.9	6.6	10.6	15.6	18.3	22.2	25.2	25.3	22.9	17.7	12.8	6.3
		MCDB	6.4	10.2	15.3	21.6	23.8	26.4	30.1	29.8	26.3	20.5	15.7	8.9
	5%	WB	2.2	4.6	8.7	13.9	17.3	21.3	24.6	24.7	21.8	16.4	11.3	4.8
		MCDB	4.6	8.2	14.1	19.9	22.8	25.4	29.2	29.0	24.5	19.6	14.4	7.5
	10%	WB	0.9	3.1	7.2	12.5	16.2	20.5	24.0	24.2	20.7	15.1	9.8	3.4
		MCDB	3.2	6.5	12.1	17.9	21.7	24.7	28.2	28.3	23.7	18.7	12.9	6.0
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.2	10.8	12.1	11.7	9.6	7.3	7.6	9.0	10.8	9.8	8.3
		MCDBR	10.3	12.8	15.3	16.5	15.4	12.6	9.7	10.2	10.9	12.7	13.1	11.4
	5% WB	MCWBR	7.0	8.2	8.4	7.7	6.3	4.8	3.7	3.7	5.0	6.3	7.7	7.7
		MCDWR	9.2	11.5	13.0	13.4	11.9	9.1	8.5	8.3	8.2	9.5	10.8	10.6
Clear-Sky Solar Irradiance	taub		0.400	0.444	0.523	0.555	0.564	0.623	0.579	0.536	0.466	0.443	0.423	0.396
		taud	2.036	1.959	1.800	1.781	1.800	1.734	1.911	2.022	2.162	2.096	2.068	2.069
	Ebn at Noon		754	767	741	746	746	701	730	753	788	762	723	728
		Edh at Noon	129	158	202	217	216	231	192	169	139	136	125	118
All-Sky Solar Radiation	RadAvg		2.60	3.33	4.33	4.96	5.57	5.06	4.29	4.36	3.95	3.59	2.61	2.32
	RadStd		0.15	0.29	0.42	0.39	0.51	0.41	0.58	0.60	0.44	0.30	0.31	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	-1.30	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	N/A

Nomenclature: See separate page